PUSHFORWARD ALONG A FINITE ETALE MORPHISM PRESERVES LOCALLY CONSTANT SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the reduction to the local model is written out, the model case is closed by the direct observation about finite products, and the closing 'This is clear' and 'Similarly' only repeat that verification for abelian groups and for modules.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Locally constant sheaves of sets, abelian groups and Lambda-modules on an etale site. Same chapter, the notion of a (finite) locally constant sheaf on X_etale and its behaviour under products. Supplies the notion the lemma asserts is preserved by the pushforward.
- Local structure of a finite etale morphism. Etale Morphisms, Lemma etale-lemma-finite-etale-etale-local: a finite etale morphism is etale locally a disjoint union of isomorphisms. Supplies the local model to which the verification is reduced.
- Pushforward commutes with etale localization. Same chapter, the compatibility of the small pushforward with base change along an etale morphism. Allows the assertion to be checked on the local model.

Source reference. Original lemma 04HN: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9986–9986. Navigation pointer invoked at proof line 9986; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : X \to Y$ be a finite étale morphism of schemes. If $\mathcal{F}$ is a (finite) locally constant sheaf of sets, (finite) locally constant sheaf of abelian groups, or (finite type) locally constant sheaf of Λ-modules on $X_{étale}$, the same is true for $f_*\mathcal{F}$ on $Y_{étale}$.*

Proof. The construction of f_* commutes with étale localization. A finite étale morphism is locally isomorphic to a disjoint union of isomorphisms, see Étale Morphisms, Lemma 04HN. Thus the lemma says that if $\mathcal{F}_i$, $i = 1, \ldots, n$ are (finite) locally constant sheaves of sets, then $\prod_{i=1,\ldots,n} \mathcal{F}_i$ is too. This is clear. Similarly for sheaves of abelian groups and modules. □